

Remainder topics

Intro to Robotics

1 Torques and Forces on Joints

Two type of joints -revolute and prismatic.

Revolute produces torque and prismatic produces force.

Equation: $\tau = {}^0J\hat{F}$ where,;

τ : is the vector of torques and forces for each joint

$\hat{F}$: the unit forces

Example

$${}^0J = \begin{bmatrix} -s_1 d_3 c_2 & -c_1 s_2 d_3 & c_1 c_2 \\ -c_1 d_3 c_2 & -s_1 s_2 d_3 & s_1 c_2 \\ 0 & c_2 d_3 & s_2 \end{bmatrix}$$

$\tau = ({}^0J)^T \cdot \hat{F}$

$$\begin{bmatrix} \tau_1 \\ \tau_2 \\ F_3 \end{bmatrix} = \begin{bmatrix} -s_1 d_3 c_2 & -c_1 s_2 d_3 & c_1 c_2 \\ -c_1 d_3 c_2 & -s_1 s_2 d_3 & s_1 c_2 \\ 0 & c_2 d_3 & s_2 \end{bmatrix}^T \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix}$$

2 Lagrangian Mechanics (Torques and Forces)

Goal: Find the torques or forces acting on a joint

$\mathcal{L} = \text{kinetic(KE)} - \text{potential(PE)}$

- $\text{KE} = \frac{1}{2}mv^2$
- $\text{PE} = mgh$
- $\theta = \text{position}$
- $\dot{\theta} = \text{velocity}$
- $R \rightarrow \text{torque (revolute)}$

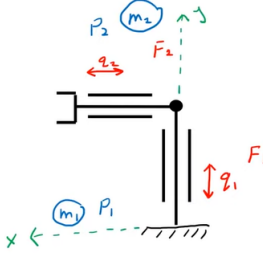
- $P \rightarrow$ force (prismatic)

2.1 Algorithm

- Robot manipulator $\rightarrow$ Position vector (θ)
- Differentiate $\rightarrow$ velocity vector ($\dot{\theta}$)
- $KE = \frac{1}{2}mv$, $PE = mgh$
- $\mathcal{L} = KE - PE$
- $\tau = \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) - \frac{\partial \mathcal{L}}{\partial \theta} - [\text{external forces/torques}] \theta$ or $\mathbf{F}$

2.2 Example

Worked Example



- Two prismatic joints - P1 and P2
- Masses of links - m1 and m2
- Displacement of joints - q1 and q2
- Input forces produced by actuators - F1 and F2

Position vectors

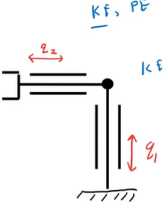
$$P_{m1} = \begin{bmatrix} 0 \\ q_1 \end{bmatrix} \quad P_{m2} = \begin{bmatrix} q_2 \\ q_1 \end{bmatrix}$$

Velocity vectors

$$V_{m1} = \begin{bmatrix} 0 \\ \dot{q}_1 \end{bmatrix} \quad V_{m2} = \begin{bmatrix} \dot{q}_2 \\ \dot{q}_1 \end{bmatrix}$$

$V_m = \frac{dP_m}{dt}$

Worked Example



KE, PE

$$KE = \frac{1}{2}mv^2 = \frac{1}{2}m v_x^2 + \frac{1}{2}m v_y^2$$

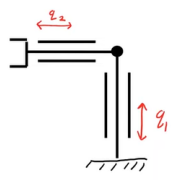
$$KE = \frac{1}{2}m (v_x^2 + v_y^2)$$

$$KE_{m1} = \frac{1}{2}m_1(\dot{q}_1^2) \quad ; \quad KE_{m2} = \frac{1}{2}m_2(\dot{q}_2^2 + \dot{q}_1^2)$$

PE = mgh $\leftarrow$ height (vertical component)

$$PE_{m1} = m_1 g q_1 \quad , \quad PE_{m2} = m_2 g q_1$$

Worked Example



$$\mathcal{L} = K\dot{\mathbf{E}} - P\dot{\mathbf{E}}$$

$$\mathcal{L} = K\dot{\mathbf{E}}_{m_1} + K\dot{\mathbf{E}}_{m_2} - P\dot{\mathbf{E}}_{m_1} - P\dot{\mathbf{E}}_{m_2}$$

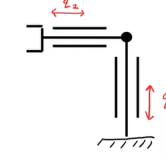
$$\mathbf{p}_{m_1} = \begin{bmatrix} 0 \\ q_1 \end{bmatrix} \quad \mathbf{p}_{m_2} = \begin{bmatrix} q_2 \\ q_1 \end{bmatrix}$$

$$\mathbf{v}_{m_1} = \begin{bmatrix} 0 \\ \dot{q}_1 \end{bmatrix} \quad \mathbf{v}_{m_2} = \begin{bmatrix} \dot{q}_2 \\ \dot{q}_1 \end{bmatrix}$$

$$\mathcal{L} = \frac{1}{2} m_1 \dot{q}_1^2 + \frac{1}{2} m_2 (\dot{q}_2^2 + \dot{q}_1^2) - m_1 g q_1 - m_2 g q_1$$

$$\mathcal{L} = m_1 \left(\frac{1}{2} \dot{q}_1^2 - g q_1 \right) + m_2 \left(\frac{1}{2} \dot{q}_2^2 + \frac{1}{2} \dot{q}_1^2 - g q_1 \right)$$

Worked Example



Prismatic
 F_{m_1}, F_{m_2}

$$\mathcal{T} = \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) - \frac{\partial \mathcal{L}}{\partial \theta} - \left[\text{External Forces / Torques} \right]_0$$

$$F_{m_1} = \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_1} \right) - \frac{\partial \mathcal{L}}{\partial q_1} - F_1(q_1)$$

$$F_{m_2} = \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_2} \right) - \frac{\partial \mathcal{L}}{\partial q_2} - F_2(q_2)$$

$$\frac{\partial \mathcal{L}}{\partial q_1} = m_1 \dot{q}_1 + m_2 \dot{q}_1$$

$$\frac{\partial \mathcal{L}}{\partial q_1} = -m_1 g - m_2 g$$

$$\frac{\partial \mathcal{L}}{\partial q_2} = m_2 \dot{q}_2$$

$$\frac{\partial \mathcal{L}}{\partial q_2} = 0$$

$$F_{m_1} = \frac{d}{dt} (m_1 \dot{q}_1 + m_2 \dot{q}_1) + m_1 g + m_2 g - F_1(q_1)$$

$$F_{m_2} = \frac{d}{dt} (m_2 \dot{q}_2) - 0 - F_2(q_2)$$

$m_2 \ddot{q}_2 \leftarrow \text{acc}$

3 Trajectory Planning and Generation

Trajectory: Time history of position, velocity and acceleration for each DOF.

- Run time
- Path update rate (60-2000hz)
- You also want smooth motion(minimize wear and tear)
- **Via points:** auxiliary/intermediate points between initial and final positions
- Optimization problem because there are many ways to potentially go from initial to final

Path Generation Methods

- Joint Space Schemes $\rightarrow$ using function of joint angles(in terms of θ), each via point(including final and initial) is converted to a set of joint angles using inverse kinematics

- Cartesian Schemes $\rightarrow$ directly uses the position and orientation of the robot frames along with the path points as specified by the user to generate the trajectory.

In general, joint space schemes are easier to use and compute. This is because cartesian schemes require inverse kinematics to be used at every single point along the path, in the path update. So higher computational burden on the robot.

3.1 Joint Space Schemes

Is split into two sub methods:

- Cubic Polynomials(can be even higher order)
- Parabolic Blends

3.2 Cubic Polynomials

The idea here is to find an equation(3rd degree polynomial) that satisfies the following constraints:

- $\theta(t_0) = \theta_0$ aka the initial position at time 0 is the initial position of the robot
- $\dot{\theta}(0) = 0$ aka the derivative of the initial position has velocity 0
- $\theta(t_f) = \theta_f$ the final position at time f, is at the intended final position

To model a polynomial with 4 constraints you need at least a 3rd degree polynomial. If you add via points, then you can increase the degree of the polynomial by adding additional constraints.

Form of functions:

- $\theta(t) = a_0 + a_1t + a_2t^2 + a_3t^3$ Position
- $\dot{\theta}(t) = a_1 + 2a_2t + 3a_3t^2$ Velocity
- $\ddot{\theta}(t) = 2a_2 + 6a_3t$ Acceleration

Example:

Cubic Polynomials - Example

Initial conditions at $t=0$:

- $\theta(0) = \theta_0 = 2^\circ$
- $\dot{\theta}(0) = 0$
- $\theta(0) = \theta_0 = 15^\circ$
- $\dot{\theta}(0) = 0$
- $\theta(0) = \theta_0 = 20^\circ$
- $\dot{\theta}(0) = 0$

Final conditions at $t=2$:

- $\theta(2) = \theta_f = 2^\circ$
- $\dot{\theta}(2) = 0$
- $\theta(2) = \theta_f = 15^\circ$
- $\dot{\theta}(2) = 0$
- $\theta(2) = \theta_f = 20^\circ$
- $\dot{\theta}(2) = 0$

2 Functions $\rightarrow$ $\theta_B(t:0 \rightarrow 2)$ $\theta_C(t:0 \rightarrow 2)$

Position:

$$\theta(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3$$

$$\dot{\theta}(t) = a_1 + 2a_2 t + 3a_3 t^2$$

$$\ddot{\theta}(t) = 2a_2 + 6a_3 t$$

From $A \rightarrow B$:

$$\theta(0) = \theta_0 = a_0 + a_1(0) + a_2(0^2) + a_3(0^3) = 2 \rightarrow a_0 = 2$$

$$\dot{\theta}(0) = 0 = a_1 + 2a_2(0) + 3a_3(0^2) \rightarrow a_1 = 0$$

$$\theta(2) = 2 = a_0 + a_1(2) + a_2(2^2) + a_3(2^3) = 2 + 4a_2 + 8a_3 = 2 \rightarrow 4a_2 + 8a_3 = 0$$

$$\dot{\theta}(2) = 0 = a_1 + 2a_2(2) + 3a_3(2^2) = 4a_2 + 12a_3 = 0$$

Solving the system:

$$\begin{cases} 4a_2 + 8a_3 = 0 \\ 4a_2 + 12a_3 = 0 \end{cases} \Rightarrow a_2 = 9.75, a_3 = -3.25$$

Final polynomial for $A \rightarrow B$:

$$\theta(t) = 2 + 9.75t^2 - 3.25t^3$$

Final polynomial for $B \rightarrow C$:

$$\theta(t) = 15 + 3.75t^2 - 1.25t^3$$

3.3 Parabolic Blends

To solve discontinuous velocities, we use parabolic blends.

Explanation:

Parabolic Blends

Problem: discontinuous velocities

blend: constant acceleration

$t_b = \text{blend1} = \text{blend2}$

Path Trajectories

$$\theta_b = \theta_0 + \frac{1}{2} \ddot{\theta} t_b^2 \quad \text{joint 1}$$

$$t_b = \frac{t}{2} - \frac{\sqrt{\ddot{\theta}^2 t^2 - 4\ddot{\theta}(\theta_f - \theta_0)}}{2\ddot{\theta}}$$

$$\ddot{\theta} \geq \frac{4(\theta_f - \theta_0)}{t^2}$$

Path Trajectories

$$\theta(t) = \theta_0 + \frac{1}{2} \ddot{\theta} t^2 ; 0 \leq t < t_b \rightarrow \text{blend 1}$$

$$\theta(t) = \theta_b + \ddot{\theta} t_b (t - t_b) ; t_b \leq t < t_f - t_b \rightarrow \text{linear}$$

$$\theta(t) = \theta_f - \frac{1}{2} \ddot{\theta} (t - t_f)^2 ; t_f - t_b \leq t \leq t_f \rightarrow \text{blend 2}$$

Example:

Parabolic Blends - Example

A $\xrightarrow{t=3}$ B

$\theta_a = 15^\circ$ $\theta_b = 75^\circ$

$\dot{\theta}_a = 0$ $\dot{\theta}_b = 0$

$\ddot{\theta} = 30^\circ/s^2$

$\theta_b = \theta_a + \frac{1}{2} \ddot{\theta} t_b^2$
 $t_b = \frac{t}{2} - \frac{\sqrt{\ddot{\theta} t^2 - 4\ddot{\theta}(\theta_b - \theta_a)}}{2\ddot{\theta}}$
 $\ddot{\theta} \geq \frac{4(\theta_b - \theta_a)}{t^2}$

$\ddot{\theta} > \frac{4(75-15)}{3^2} = 26.7^\circ/s^2$ $\ddot{\theta} = 30^\circ/s^2$

$t_b = \frac{3}{2} - \frac{\sqrt{30^2(3^2) - 4(30)(75-15)}}{2(30)} = 1.0 \text{ s} \leftarrow$

$\theta_b = 15 + \frac{1}{2}(30)(1^2) = 30^\circ \leftarrow$

blend 1 $\rightarrow \theta(t) = 15 + 15t^2$; $0 \leq t < 1$

linear $\rightarrow \theta(t) = 30 + 30(t-1)$; $1 \leq t < 2$

blend 2 $\rightarrow \theta(t) = 75 - 15(t-2)^2$; $2 \leq t \leq 3$

} Path trajectory

Assumption

Blend times for both sections are equal

↓

Velocity at end of blend = velocity at linear section

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4 Credit

These are my notes from the robotics playlist youtube channel @thatsengineering5235, all of the external images used are examples from their videos.